

## **Batch Flow: A POS BASED SUPPLY CHAIN MANAGEMENT SYSTEM**

*"Smart POS. Smarter Supply Chain."*



**A Collaborative Project for Web Systems and Technologies and Management Information Systems Subjects**

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## **CHAPTER 1 — INTRODUCTION**

### **Background of the Study**

Many bakeries today still rely on manual methods in managing sales, inventory, and product supplies, which can lead to delays, inaccurate records, and product shortages. As customer demand continues to increase, bakery businesses need a more efficient and organized system to monitor daily transactions and supply chain operations. This study focuses on developing a web-based POS and supply chain management system that can help bakeries improve operational efficiency, reduce errors, and provide better service to customers.

### **Problem Statement**

There are a lot of factors at play here, lack of proper communication between subordinates and higher ups can result in some system proposal getting canceled due to not accepting the system. Others are can be because people who work there are not into the technology. Bakeries are getting less of a recognition because of newer more popular bakeshops like Goldilocks or the “oven on wheels” style of marketing made bakeshops essentially getting less recognition.

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### **Objectives of the System**

#### **General Objective**

To develop a Batch Flow: A POS-Based Supply Chain Management System for efficient bakery sales, inventory, and supply management. Specific Objectives

#### **Specific Objective:**

- To record and manage sales transactions accurately.
  - To monitor inventory and supply stocks efficiently.
  - To track supplier orders and deliveries.
  - To reduce manual errors through automation.
  - To generate sales and inventory reports.
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### **Scope and Limitations**

#### **Scope**

It is designed for bakery businesses to manage sales transactions, inventory, and supply chain operations efficiently. The system includes features such as point-of-sale transactions, inventory monitoring, supplier and delivery tracking, and report generation for sales and stock management. It is intended to help bakery staff and administrators improve operational efficiency, accuracy, and record management.

## **Limitations**

- Limited only to bakery-related operations.
  - Requires an internet connection.
  - No advance accounting.
  - No online payment integration
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## **Significance of the Study**

For bakery owners and staff, the system provides a more convenient and reliable way of handling daily business activities. Customers may also benefit from faster and more accurate service during transactions. Furthermore, this study can serve as a reference for future researchers who plan to develop similar business management systems.

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## **Target Users**

Firstly we want to target the bakery owners, cashiers, inventory staff, customers, and system administrators is a great candidate for it allowing them to edit and track transaction and data records. For the employees we wanted for them to give a simple POS function that is easy to use and allows for simpler tasks.

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## CHAPTER 2 — RELATED LITERATURE & SYSTEMS

### Related Literature

Liew, C. C., & Alduais, N. (2022) said that their study focused on developing a web-based inventory and customer management system for a retail bakery to improve the efficiency of sales, inventory tracking, and customer transactions. The researchers concluded that the system helped reduce manual errors, improved inventory monitoring, and provided faster and more organized bakery operations.

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### Related Systems

#### Toast POS

##### Features:

- Point-of-Sale (POS) transaction processing.
- Inventory and menu management.
- Online ordering and delivery integration.
- Sales and analytics reporting.
- Supplier and stock tracking.
- Kitchen display and order management system.

##### Strengths

- Easy-to-use restaurant and bakery management interface.
- Real-time inventory and sales monitoring.
- Supports online ordering and multiple payment methods.
- Provides detailed reports for business analysis.
- Scalable for growing food businesses and multiple branches.

##### Weaknesses

- Requires stable internet connection for full functionality.
  - High subscription and hardware costs for small businesses.
  - Limited flexibility because it uses proprietary payment processing.
  - Some users reported customer support and system bug issues.
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## CHAPTER 3 — SYSTEM ANALYSIS & DESIGN

### Development Methodology

Identify and explain the software development methodology used in developing the system.

##### Examples:

- **Waterfall Model**
- **Agile Methodology**

- Scrum
- Rapid Application Development (RAD)

Students must:

- Explain the chosen methodology
- Discuss each phase/process
- Explain why it is appropriate for the project

## Proposed System Features

### A. Operational Features (Web System)

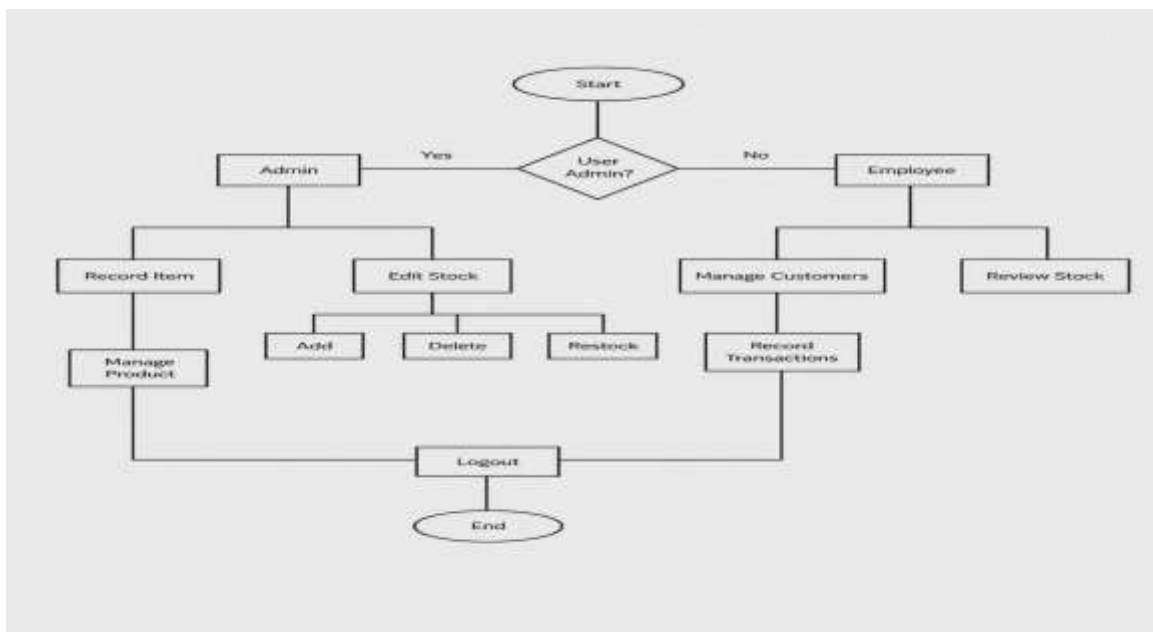
- Sales Transaction Processing.
- Inventory and Stock Monitoring.
- Product and Pricing Management.
- Supplier and Delivery Tracking.
- Order and Receipt Management.

### B. MIS Features (Management Side)

- Sales Report Generation.
- Inventory Status Reporting.
- Dashboard Analytics and Business Monitoring.

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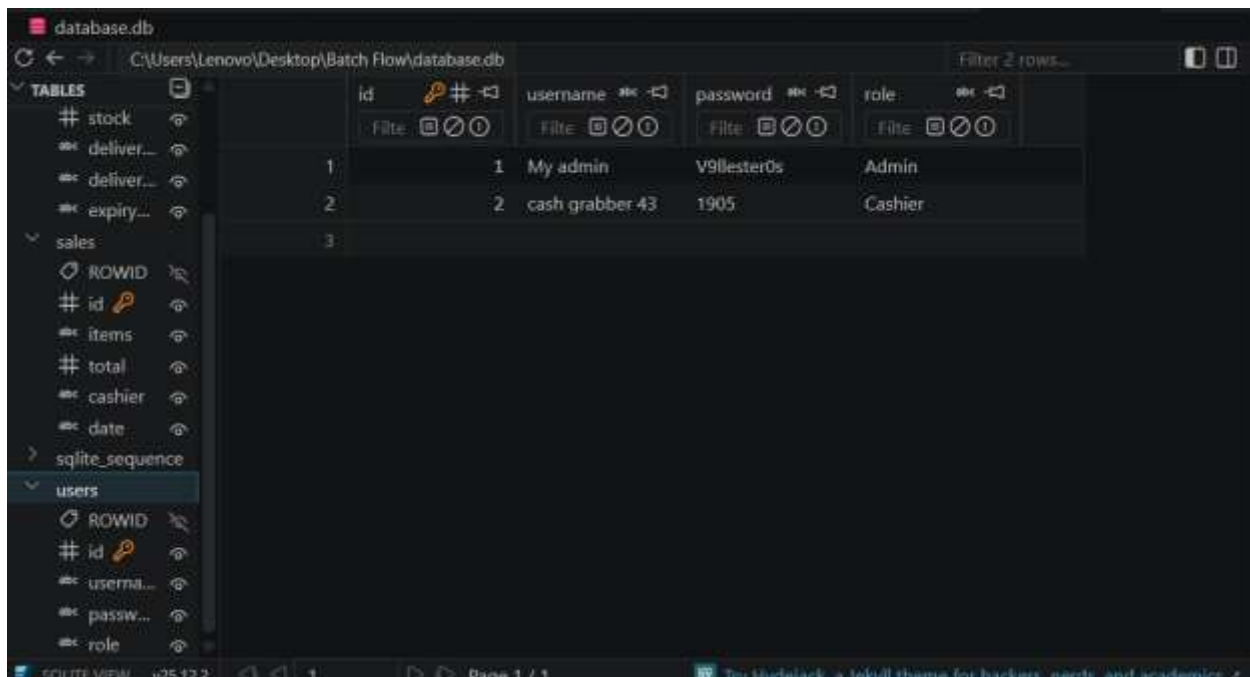
## System Overview Diagram



## Database Design

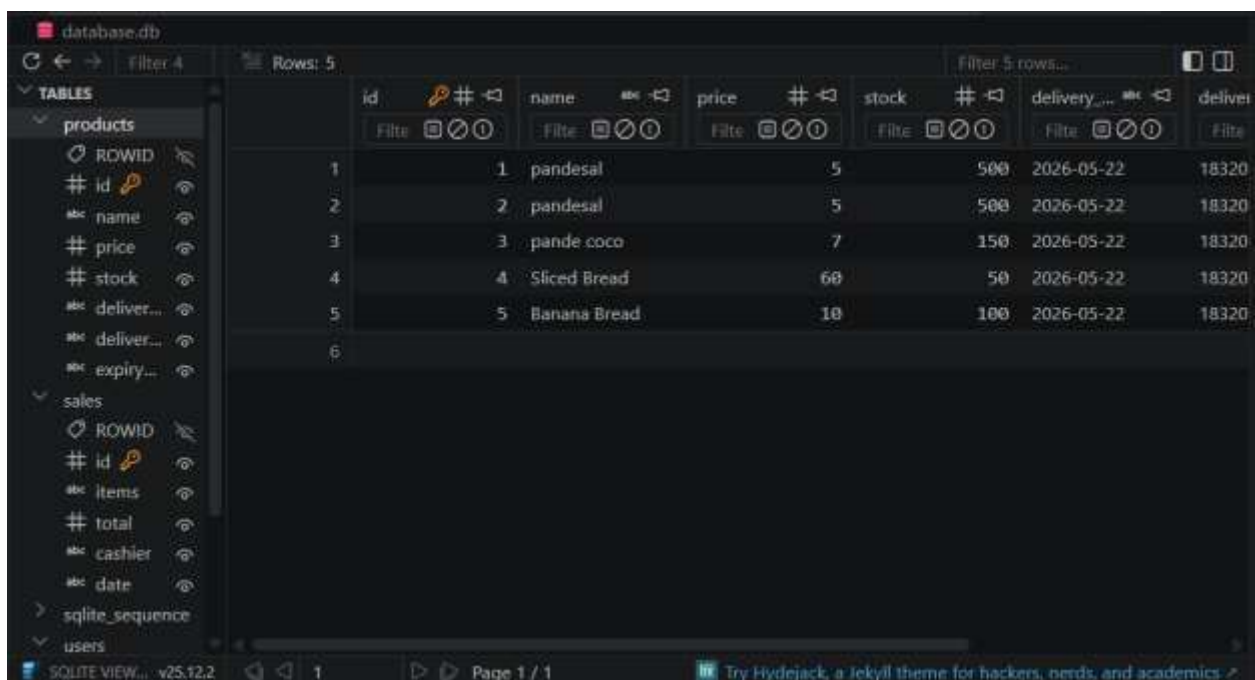
Include:

User database:



The screenshot shows a SQLite database viewer with the file path 'C:\Users\Lenovo\Desktop\Batch Flow\database.db'. The 'users' table is selected in the left sidebar. The table has five columns: 'id', 'username', 'password', and 'role'. The 'id' column is the primary key. There are two rows of data.

| id | username        | password    | role    |
|----|-----------------|-------------|---------|
| 1  | My admin        | V9llester0s | Admin   |
| 2  | cash grabber 43 | 1905        | Cashier |



The screenshot shows the same SQLite database viewer with the 'products' table selected. The table has seven columns: 'id', 'name', 'price', 'stock', 'delivery\_date', and 'delivery\_price'. The 'id' column is the primary key. There are five rows of data.

| id | name         | price | stock | delivery_date | delivery_price |
|----|--------------|-------|-------|---------------|----------------|
| 1  | pandesal     | 5     | 500   | 2026-05-22    | 18320          |
| 2  | pandesal     | 5     | 500   | 2026-05-22    | 18320          |
| 3  | pande coco   | 7     | 150   | 2026-05-22    | 18320          |
| 4  | Sliced Bread | 60    | 50    | 2026-05-22    | 18320          |
| 5  | Banana Bread | 10    | 100   | 2026-05-22    | 18320          |



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### Functional Requirements

- Allow users to log in and log out securely.
  - Record and process sales transactions through the POS module.
  - Manage bakery products, prices, and categories.
  - Monitor inventory levels and ingredient stocks in real time.
  - Track supplier information, orders, and deliveries.
  - Generate sales, inventory, and supply reports.
  - Provide stock alerts for low inventory items.
  - Store customer transaction history and order records.
  - Allow administrators to manage user accounts and access permissions.
  - Backup and retrieve system data when needed.
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### Non-Functional Requirements

- Performance
  - Security
  - Reliability
  - Usability
  - Scalability
  - Compatibility
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## CHAPTER 4 — SYSTEM DEVELOPMENT & TESTING

- HTML
- CSS
- JavaScript
- MySQL

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## **System Implementation**

How the system was developed:

Batch Flow: A POS-Based Supply Chain Management System was developed as a web-based platform to help bakery businesses manage sales, inventory, and supply chain operations efficiently. The system was created using web development technologies such as HTML, CSS, JavaScript, and MySQL to provide a user-friendly and functional interface for bakery staff and administrators. The development focused on automating manual bakery processes, improving transaction accuracy, and providing organized data management.

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## **Testing and Evaluation**

**Development Process:**

- Planning Phase
- Requirement Analysis
- System Design
- System Development
- Testing Phase
- Implementation Phase

**Deployment Process:**

- The system database and source code were uploaded to a web server or hosting platform.
- User accounts and system settings were configured for bakery staff and administrators.
- Initial bakery products, inventory records, and supplier information were added to the database.
- Users were trained on how to use the POS, inventory, and reporting features of the system.
- The system was deployed for daily bakery operations and monitored regularly for maintenance and updates.



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## **CHAPTER 5 — CONCLUSION & RECOMMENDATION**

### **Conclusion:**

Batch Flow: A POS-Based Supply Chain Management System successfully improved bakery operations by automating sales, inventory, and supply management processes. The system solved problems related to manual recording, slow transactions, and inaccurate inventory tracking. It also provided benefits such as faster transactions, organized data management, real-time stock monitoring, and improved business efficiency.

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### **Recommendations**

#### **Suggest:**

For future improvements, the system may include a mobile application version for easier access and monitoring using smartphones and tablets. Additional features such as online ordering, online payment integration, barcode scanning, and customer loyalty programs can also be added to improve customer service and business operations. Possible upgrades may include cloud-based storage, advanced sales analytics, automated supplier ordering, and multi-branch management to support the future growth of the bakery business.

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## **APPENDICES**

### **Appendices**

#### **Include:**

- **Survey Forms**
  - **Interview Questions**
  - **User Manual**
  - **Documentation Photos**
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### **References**

**Use proper citation format:**

- **APA**